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YellowScan Toolkit User Manual.

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About



The YellowScan Toolkit (referred to as **ytk** in this document) is an advanced Command Line Interface (CLI) version of the YellowScan CloudStation software. It allows all processes available in CloudStation with no Graphical User Interface (GUI) with the possibility of being batchable.

The use of **ytk** is **destined to advanced and experimented users**. It contains no guardrail or check on whether something makes sense to do or not in your data. Inconsiderate processing can lead to bad, unusable or incorrect data. User discretion is advised.

Installation

Hardware Requirement

	Minimal Configuration	Mid-range Configuration	Recommended Configuration
<i>CPU</i>	Intel or AMD 4C/8T CPU with SSE4.2 support	Intel or AMD 6C/12T CPI (ex : Intel 8700 or AMD 5600X)	Intel or AMD – High core count, high frequency (ex: Intel 13800k, AMD 7900X)
<i>GPU</i>	None	Dedicated Nvidia GPU with at least 2GB dedicated VRAM	Dedicated Nvidia GPU with at least 4GB dedicated VRAM (ie: Nvidia RTX 3060)
<i>RAM</i>	8 GB	16 GB	1 GB / CPU thread (ie: 24 threads CPU = at least 24 GB RAM)
<i>Storage</i>	500 MB for installation	1 TB SSD	TB NVMe SSD

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Setup



The YellowScan Toolkit is currently distributed for Windows 10 and Windows 11. The software is not available on Linux nor MacOS.

It is bundled with the CloudStation GUI installer and requires a CloudStation Ultimate license to activate. You can find the latest version of CloudStation here:

<https://software.yellowscan.com/cloudstation/latest/win64>

The installation is straightforward, follow the install wizard to the last step. You will have to agree to the End User License Agreement to proceed with the installation.

Once the installation is completed, you can start the YellowScan Toolkit from the command line by calling the command **ytk**.

By default, the executable is located in **C:\Program Files\YellowScan\CloudStation\bin\ytk.exe**

Licensing

The first time you run ytk, you'll notice very few options are available. That's because the toolkit requires a license. Before activation, two TCP ports need to be opened in your firewall(s) to be able to access the license server:

- Port 5053
- The second port is unique to your license

To import a new license, use the **license** subcommand:

```
ytk license --set C:\path\to\license.key
```

To check the status of your license and the modules available to you, run the following command:

```
ytk license --info
```

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Working Directories



The YellowScan Toolkit can generate a lot of data to store your project. To keep your computer clean, all the data necessary for a project is stored in a working directory. Since the toolkit requires one **workdir** per project, it is best to group them all in a directory to keep track of them.

Almost all the subcommands will require you to point to a working directory, starting with the “extraction” subcommand.

Creating the working directory

The working directory is not created by the toolkit. You must create it manually first:

```
mkdir C:\myWorkdirs
```

Warning: It is strongly recommended to use empty directories as workdirs since the toolkit can and will delete unknown content in the directories it is given as workdirs.

Commands & Subcommands

Help

To see the list of subcommands and a short description of what they do, use the **--help / -h** flag:

```
ytk -h
```

Each subcommand has its own list of arguments. To see the list of arguments for a subcommand, you can also use the **-h** flag. For example, to get help on the **license** command, type:

```
ytk license -h
```

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Command Usage



ytk is used by giving a subcommand, followed by the list of parameters required by this subcommand:

```
ytk <subcommand> <parameters>
```

For example, the **extraction** command looks like this:

```
ytk extraction --extract C:\path\to\workdir --ins C:\path\to\trajectory.js --  
timestamps auto
```

It is possible to chain subcommands one after another to execute their functions sequentially.

```
ytk <subcommand> <parameters> <subcommand> <parameters>
```

INI Files

However, chaining long subcommands gets complex to both write and keep track of everything being correct. The YellowScan Toolkit can read and execute commands from an .ini file using the **--config / -c** parameter

```
ytk -c C:\path\to\config.ini
```

Configuration files are .ini files. Each command is a block with the parameters listed after. Here's an example for the extraction command:

```
; You can add comments in your file by starting a line with the ";" character  
; Subcommands are enclosed in square brackets.  
; Parameters are attached to the subcommand that precedes them  
; Parameters do not start with "--"  
  
[extraction]  
extract=C:\path\to\workdir  
ins=C:\path\to\trajectory.js  
timestamps=auto
```

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Pre-processing



This section deals with all steps that can or should be done before extracting data.

SBET Generation

Generating proper final data will require a SBET to be generated. You can either use the GUI from your trajectory processing software or the CLI for it. Both SBG Systems Qinertia and Applanix POSPac support command line instructions and can be called from your preferred terminal or in a script.

Please refer to their respective User Manuals for proper documentation.

Fullwave

When using a YellowScan Navigator bathymetric LiDAR system, the recorded data is a full wave form .yswf file. You need to convert it to a .yse file by detecting and extracting echoes from the raw file. You can choose to extract for all timestamps, or only your strips if you have existing timestamps. Note that extracting all timestamps is not necessary and will take significantly longer.

The detection sensibility has the same presets as in CloudStation:

- Low: 0
- Medium: 1
- High: 2

```
ytk fullwave --ins-data C:\path\to\flight1\trajectory.yse --data-file  
C:\path\to\flight1\rawdata.yswf -t auto --detection-sensibility 1
```

RiMTA

To process and generate YellowScan Voyager-Series data, you will need to solve for MTAs using RiMTA. This will convert your .rxp data into MTA-solved .SDCX file that will then be usable later. The newly created file is in the same folder as the input file.

```
ytk rimta --rxp C:\path\to\flight1\trajectory.rxp
```

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Creating a cs.cache



This section will show you how to extract data and create a cs.cache, to then be able to manipulate, refine and export your data. It is necessary to have one prior to doing anything with your data. A cs.cache is YellowScan's internal data format regarding all things processing. Do not tamper with a cs.cache! They will always be created at a workdir's root.

Extraction

The subcommand to create a project is **extraction**. It takes an empty **workdir** as parameter. Provide the INS (.ys) path and optionally the SBET and LiDAR profile to start an extraction. By default, the cs.cache folder is created where the INS file is located, but you can specify a different directory for it. However, you will need to have the cs.cache to reopen the project.

```
1 mkdir C:\myWorkdirs\demo-project
2 ytk extraction --extract C:\myWorkdirs\demo-project --ins
  C:\path\to\flight1\trajectory.ys --sbet C:\path\to\flight1\trajectory_sbet.txt
  --profile C:\path\to\flight1\lidar-calibration.profile --timestamps auto
```

Override, Preserve, Append

If your project is made of multiple flights, you can call extraction multiple times on the same **workdir** using the following flag: **--append** here's an example for a full extraction and export in LAZ 1.4 format:

```
1 mkdir C:\myWorkdirs\demo-project
2 ytk extraction --extract C:\myWorkdirs\demo-project --ins
  C:\path\to\flight1\trajectory.ys --sbet C:\path\to\flight1\trajectory_sbet.txt
  --profile C:\path\to\flight1\lidar-calibration.profile --timestamps auto
3 ytk extraction --append C:\myWorkdis\demo-project --ins :\path\to\flight2\trajectory.ys
  -t auto
4 ytk export --open C:\myWorkdirs\demo-project --point-cloud-format Laz14 --proj-code
  EPSG:2454 --merge-strategy MergeFlights
```

Warning: Never add a new flight to an existing project on which you have already run processes such as strip-adjustment or classification. This will lower the quality of your data.

Extracting a project with **--override** will destroy any previously extracted data, with no confirmation. On the flip side, if you're not sure about the extraction status of a **workdir**, or want to resume, using **--preserve** will extract or reopen accordingly.

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Using a cs.cache



Once you have a cs.cache, you can now refine and enrich your data as well as export it. Almost all the other subcommands will have a **--open** parameter that will accept an existing **workdir** on which the command will work.

For all commands, use the command name followed by **-h/--help** for more information about the command.

Data Refinement Commands

Available commands:

Command	Function
strip-adjustment	Robust / Precise strip adjustment and exports PDF report.
water-classification	Water classification for bathymetric point cloud.
water-refraction	Correct georeferencing with refraction for under water points.

Warning: it is mandatory to run water-classification to run water-refraction. This should be done right after the extraction of your data. Else, you will run any subsequent algorithm on incorrect data!

Data Enrichment Commands

Available commands:

Command	Function
classification	Ground / non-ground classification.
colorization	Colorize your point cloud with camera images.
colorization-from-orthophoto	Colorize your point cloud from an orthophoto.
cut-overlap	Remove overlap from your data based on strip proximity.

Data Export Commands

Available commands:

Command	Function
digital-model	Generate DTM, DSM, DHM and hillshade from your point cloud.
export-orthophoto	Generate and export an orthophoto.
export-pointcloud	Export your point cloud as a LAS/LAZ file.

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